



Q.1. Answer the following short questions:

(6x5=30)

- Define operations research and discuss its role in decision making.
- Discuss the methods used to solve operation research models.
- Define linear programming and discuss its properties.
- Define graphical sensitivity analysis.
- Define Simplex method. Discuss its main feature and conditions.
- Discuss the iterative nature of the Simplex method.

Solve the following questions.

(3x10=30)

Q2. Solve the following problem using the graphical method.

Maximize $Z = 80x_1 + 100x_2$

subject to

$$\begin{aligned}x_1 + 2x_2 &\leq 720 \\5x_1 + 4x_2 &\leq 1800 \\3x_1 + x_2 &\leq 900\end{aligned}$$

where

$$x_1, x_2 \geq 0$$

- Q3. A carpenter has 90, 80 and 50 running feet respectively of teak, plywood and rosewood. The product A requires 2, 1 and 1 running feet of teak, plywood and rosewood respectively. While the product B requires 1, 2 and 1 running feet of teak, plywood and rosewood respectively. If A would sell for Rs. 480 and B would sell for Rs. 400 per unit, how much of each should be made and sell in order to obtain maximum gross income out of his stock of wood?
- Give a mathematical formulation to this linear programming problem.
 - Use graphical method to solve the problem.
 - Indicate clearly the feasible solution region on the graph.

Q4. Use Simplex method to solve the following problem. Provide the tables for all iterations.

Maximize $Z = 3x_1 + 2x_2$

subject to

$$\begin{aligned}2x_1 + x_2 &\leq 40 \\x_1 + x_2 &\leq 24 \\2x_1 + 3x_2 &\leq 60\end{aligned}$$

where

$$x_1, x_2 \geq 0$$